

Simple VPC Design

Abstract

This report explains the design of a basic Virtual Private Cloud (VPC) architecture. The architecture separates public-facing resources from private resources to improve security, manageability, and scalability in a cloud environment.

Introduction

A Virtual Private Cloud (VPC) is a logically isolated network within a cloud provider. It allows organizations to deploy applications securely while maintaining control over IP addresses, routing, internet access, and resource placement.

Objectives

- Understand the basic components of a VPC.
- Design public and private subnets.
- Configure internet connectivity using an Internet Gateway.
- Demonstrate secure separation of application and database resources.

Components of the Architecture

VPC: 10.0.0.0/16 network range that contains all resources.

Public Subnet: 10.0.1.0/24 subnet for internet-facing services.

Private Subnet: 10.0.2.0/24 subnet for internal services.

Internet Gateway: Enables communication between the VPC and the internet.

Route Table: Controls how traffic flows within the network.

Web Server: Hosted in the public subnet to serve users.

Database Server: Hosted in the private subnet for enhanced security.

Architecture Design

Internet → Internet Gateway → VPC (10.0.0.0/16) → Public Subnet (Web Server) and Private Subnet (Database Server). The web server receives requests from users, while the database remains protected within the private subnet.

Working Principle

Users access the web application through the internet. Traffic passes through the Internet Gateway to the web server in the public subnet. The web server communicates with the database server located in the private subnet. Direct internet access to the database is restricted.

Advantages

- Security – Sensitive resources remain isolated.
- Scalability – Additional servers and subnets can be added.
- Reliability – Network resources can be organized efficiently.
- Flexibility – Supports different application architectures.

Applications

VPCs are commonly used for hosting websites, enterprise applications, databases, development environments, and cloud-native workloads.

Conclusion

The Simple VPC Design demonstrates how cloud resources can be organized into public and private segments. This design improves security, simplifies network management, and provides a foundation for building scalable cloud solutions.